

# ELI WAHOME KOBIA

## SOFTWARE ENGINEER | BACKEND & COMPUTATIONAL LOGIC

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### PROFESSIONAL SUMMARY

Software Engineer specializing in Computational Logic and Scalable Backend Systems. Experienced in bridging complex mathematical models with production-grade software using Python and Java Spring Boot. Proven track record in delivering automated engineering solutions and enterprise-level security modules for financial institutions.

### TECHNICAL EXPERTISE

Languages: Python (Intermediate) Java (Spring Boot) JavaScript (ES6+) SQL

Backend & Data: FastAPI PostgreSQL SQLAlchemy Firebase REST APIs

Security & Auth: RBAC JWT Auth Microservices Logic BCrypt

Tools & Frameworks: Git Docker Maven Agile/Scrum

### PROFESSIONAL EXPERIENCE

Software Developer (Contract) | [Freelance Structural Project](#) Late 2025

Technologies: Python, Pandas, Mathematical Modeling

- Developed a high-precision Python toolkit to automate Eurocode 3 (BS EN 1993) structural calculations.
- Engineered a modular validation engine reducing manual calculation errors and processing time by over 60%.
- Translated strict industry standards into executable code logic for load generation and steel member design.

IT Intern | [Diamond Trust Bank \(DTB\)](#) June 2025 – Sept 2025

Technologies: Java Spring Boot 3.5, PostgreSQL, JWT, React.js

- Designed and deployed an Enterprise Role-Based Access Control (RBAC) system for secure authentication.
- Implemented granular security using JWT and BCrypt for stateless, scalable session management.
- Built a dynamic administrative dashboard in React.js to visualize permissions and manage approval workflows.

### KEY TECHNICAL PROJECTS

SmartStock: Integrated Inventory & Personnel Management Feb 2026 – Present

Technologies: Python, PostgreSQL, SQLAlchemy

- Developing a unified system to synchronize inventory tracking with staff performance and access management.
- Architecting a relational database schema to maintain data integrity across stock levels and personnel records.
- Applying System Analysis & Design principles to automate stock replenishment alerts and role-specific dashboard views.

### EDUCATION

B.Sc. Business Information Technology | [Jomo Kenyatta University of Agriculture and Technology \(JKUAT\)](#) Expected 2026

Key Coursework: Software Engineering, Object-Oriented Analysis, Principles of Management, Artificial Intelligence, Advanced Database Systems